

# Molecular Dynamics Simulation Study on the Possible Factors Affecting Stability of ODS Steel

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**Abstract.** Oxide-dispersed-strengthened (ODS) steel has excellent mechanical, thermodynamic and radiation resistant properties, which makes it an important candidate material for high-temperature reactors applications. Radiation stability of oxide is very important to study at the atomic level. In present case molecular dynamics (MD) simulation is used to study stabilities of  $Y_2O_3$  particles in ODS steel as a function of particle sizes. The size dependence is obvious for particle sizes larger than 2.8 nm. Furthermore, the stabilities depend on charges of Y and O atoms.

## 1. Introduction

The structural materials in nuclear reactors have degradation caused by irradiation damage. Research on advanced fission and fusion reactors put forward higher requirements for the safety and economy of reactors, which lead to critical needs for materials of better performance [1,2,3,4]. So far candidate materials for reactor-core components include nickel-based alloys, austenitic stainless steels, ferrite / martensitic steels, and oxide dispersion strengthening (ODS) steels [5]. ODS steel is unique for high temperature applications due to their high creep resistances from embedded oxide particles of nanometre sizes [3,5,6,7,8]. In order to further optimize ODS steels, MD simulations are needed to study atomic scale details of the structural evolution of oxide particles under high temperatures. The thermal stabilities of particles are important for their applications in high temperature fast reactors. In present study some possible factors affecting the stability of ODS are discussed by MD simulation method.

## 2. Modeling procedure

The dispersoids in ODS steels are typically Y-Ti-O oxide. As a model material,  $Y_2O_3$  embedded Fe is simulated by using LAMMPS (Large scale Atomic Molecular Massively Parallel Simulator)[9]. The interatomic potentials are described by hybrid potentials from Hammond et al [10]. The potential is linked to Ziegler-Biersack-Littmark (ZBL) potential for short-range interactions [11]. Simulation cell has dimensions of 22nm×22nm×22nm and has one oxide particle of size 5 nm embedded at the center. Periodic boundary conditions are applied on all sides. The cell is relaxed initially by isobaric-isothermal ensemble NPT for 100ps, followed by microcanonical ensemble NVE for 100ps at 300K. Figure 1 shows the cell structure. The red color represents O atoms, and green color for Y atoms, and the purple color for Fe atoms. The cell contains 1015597 atoms including 21550 O atoms and 14086 Y atoms. The Open Visualization Tool (OVITO) [12] is used for visualization of LAMMPS output.

